

Advanced Encryption Schemes

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Outline

Recap: Lecture 3

Naor–Yung: IND-CCA1 security

Naor–Yung: IND-CCA2 security

Three security notions for PKE

Same game (guess b from $ct^* \leftarrow \mathrm{Enc}(pk, m_b)$); they differ only in the access to $\mathcal{O}_{\mathrm{Dec}}(\cdot) = \mathrm{Dec}(sk, \cdot)$:

	$\mathcal{O}_{\mathrm{Dec}}$ *before* ct^*	$\mathcal{O}_{\mathrm{Dec}}$ *after* ct^*	Adversary
IND-CPA	×	×	passive
IND-CCA1	✓	×	lunchtime
IND-CCA2	✓	✓ $ct \neq ct^*$	active

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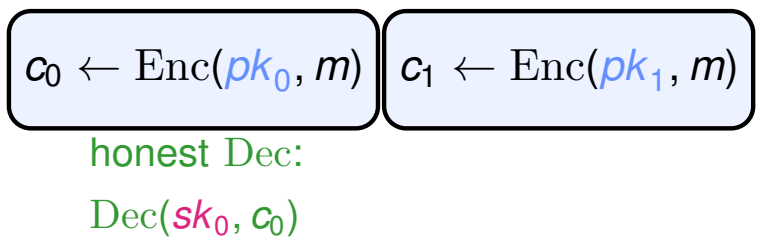
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Goal: generic transformation IND-CPA Π + "something" $\Rightarrow$ IND-CCA.
Difficulty: a reduction must answer decryption queries *without sk*.

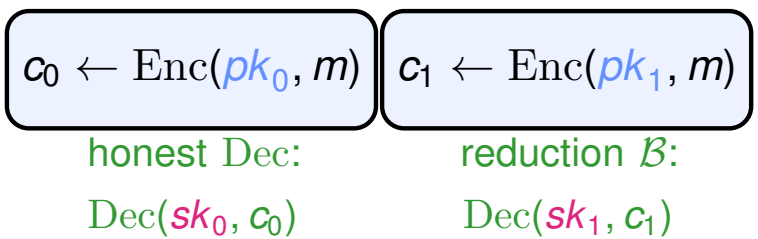
Naor–Yung: the idea

- Main idea: *Encrypt twice*, under two independent keys



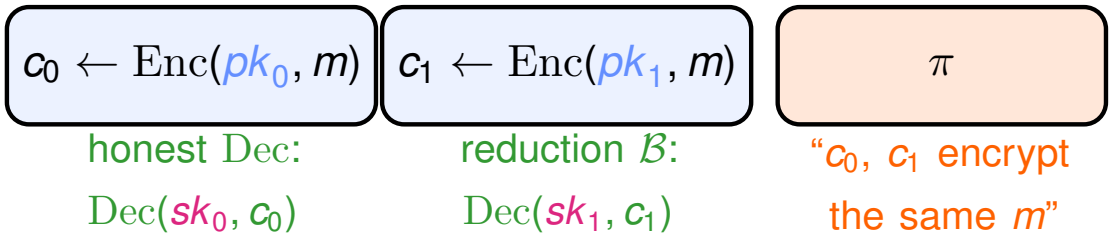
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- Problem: $\mathcal{A}$ may send **inconsistent** (c_0, c_1); then decryption with the two keys give different answers.
- Fix: attach a *non-interactive zero-knowledge proof* π that c_0, c_1 encrypt the same message; Dec rejects if π is invalid.



Naor–Yung: the construction

Building blocks: IND-CPA PKE $\Pi = (\mathrm{Gen}, \mathrm{Enc}, \mathrm{Dec})$; NIZK $(\mathrm{Setup}, \mathrm{Prove}, \mathrm{Vfy})$ for

$$\mathcal{R}_{\mathrm{NY}} = \{((pk_0, pk_1, c_0, c_1), (m, r_0, r_1)) : c_0 = \mathrm{Enc}(pk_0, m; r_0) \wedge c_1 = \mathrm{Enc}(pk_1, m; r_1)\}.$$

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$\Pi_{\mathrm{NY}} = (\Pi_{\mathrm{NY}}.\mathrm{Gen}, \Pi_{\mathrm{NY}}.\mathrm{Enc}, \Pi_{\mathrm{NY}}.\mathrm{Dec})$

$\Pi_{\mathrm{NY}}.\mathrm{Gen}(1^\lambda)$: $(pk_i, sk_i) \leftarrow \mathrm{Gen}(1^\lambda)$ for $i \in \{0,1\}$, $crs \leftarrow \mathrm{Setup}(1^\lambda)$

output $pk := (pk_0, pk_1, crs)$, $sk := sk_0$ (sk_1 erased)

$\Pi_{\mathrm{NY}}.\mathrm{Enc}(pk, m)$: $c_0 \leftarrow \mathrm{Enc}(pk_0, m; r_0)$, $c_1 \leftarrow \mathrm{Enc}(pk_1, m; r_1)$

$\pi \leftarrow \mathrm{Prove}(crs, (pk_0, pk_1, c_0, c_1), (m, r_0, r_1))$

output $ct := (c_0, c_1, \pi)$

$\Pi_{\mathrm{NY}}.\mathrm{Dec}(sk, ct)$: if $\mathrm{Vfy}(crs, (pk_0, pk_1, c_0, c_1), \pi) = 0$: output $\perp$

else output $\mathrm{Dec}(sk_0, c_0)$

NIZK in the CRS model: the three properties

$\mathrm{Setup}(1^\lambda) \to \mathit{crs},\ \mathrm{Prove}(\mathit{crs}, x, w) \to \pi,\ \mathrm{Vfy}(\mathit{crs}, x, \pi) \to \{0,1\}.$

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completeness	honest proofs of true statements verify	correctness
soundness	no PPT prover makes a **false** statement verify (honest *crs*)	consistency $\Rightarrow$ either key decrypts
zero-knowledge	$\exists$PPT simulator $\mathcal{S}$: fake (*crs*, *td*) and "real looking" proofs *without a witness* (for true statements)	π leaks nothing about *m*

IND-CCA1 security of Naor-Yung

Theorem 3.1 IND-CCA1 security of Naor–Yung [NY90]

If Π is IND-CPA secure and NIZK is complete, sound and zero-knowledge for $\mathcal{R}_{\mathrm{NY}}$, then Π_{NY} is IND-CCA1 secure.

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Game hopping proof strategy

Set $\mathrm{G}_0 \coloneqq \mathrm{Game}^{\text{IND-CCA1}}_{\Pi_{\mathrm{NY}},\mathcal{A}}(0)$ and $\mathrm{G}_6 \coloneqq \mathrm{Game}^{\text{IND-CCA1}}_{\Pi_{\mathrm{NY}},\mathcal{A}}(1)$

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- Define intermediary games $\mathrm{G}_1, \ldots, \mathrm{G}_5$ such that
  - each game differs from previous one only by a little bit
  - define $\epsilon_i \coloneqq \left| \Pr[G_{i-1} = 1] - \Pr[G_i = 1] \right|$
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  - bound each ϵ_i (via a reduction to one of the assumptions)
- Complete the proof via triangular inequality:

$$|\Pr[\mathrm{G}_0 = 1] - \Pr[\mathrm{G}_6 = 1]| \leq \sum_{i=1}^{6} |\Pr[\mathrm{G}_{i-1} = 1] - \Pr[\mathrm{G}_i = 1]| = \sum_{i=1}^{6} \epsilon_i$$

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G_0 to G_1

Differences:

- *crs* generation:
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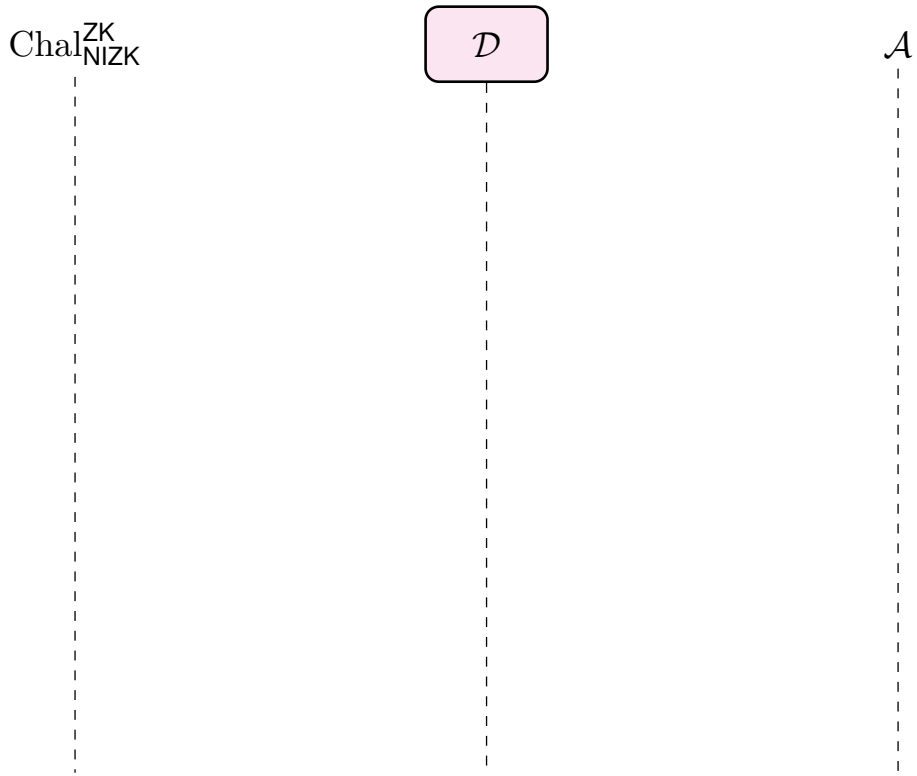
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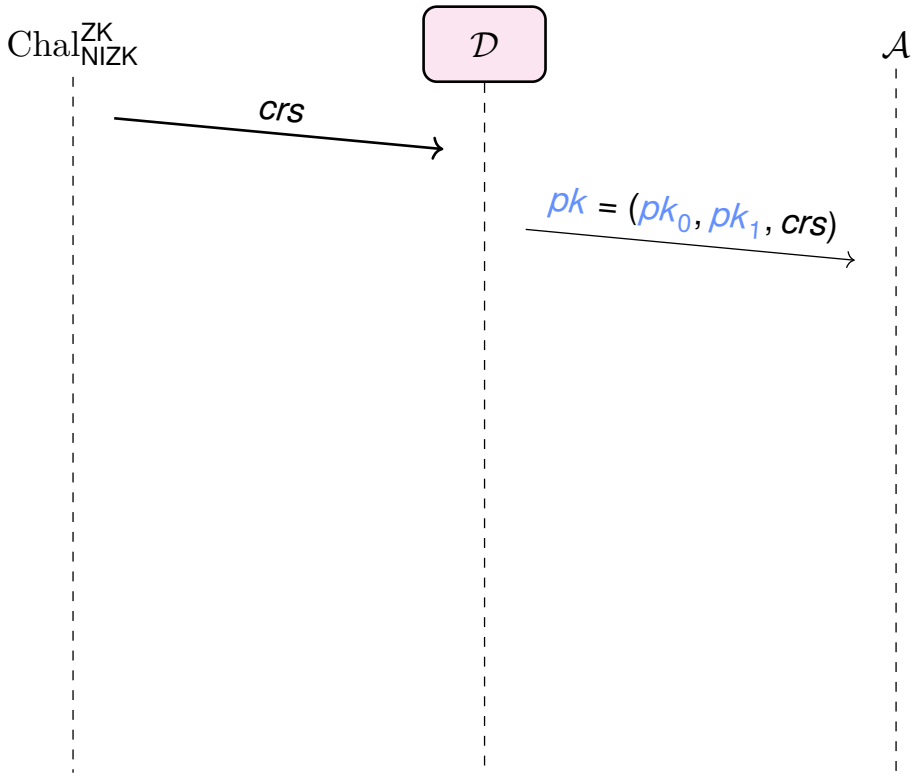
We constructed a ZK distinguisher $\mathcal{D}$:

- If $\mathcal{D}$ in the "real" experiment $\rightarrow$ simulates G_0 to $\mathcal{A}$
- If $\mathcal{D}$ in the "ideal" experiment $\rightarrow$ simulates G_1 to $\mathcal{A}$

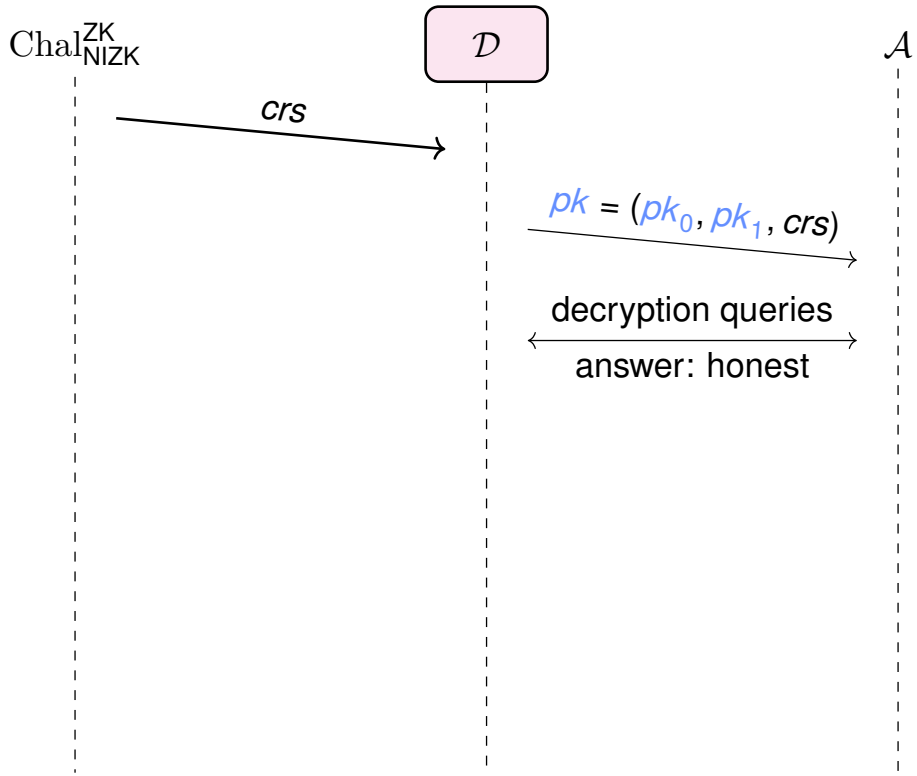
Hop $G_0 \to G_1$: the picture



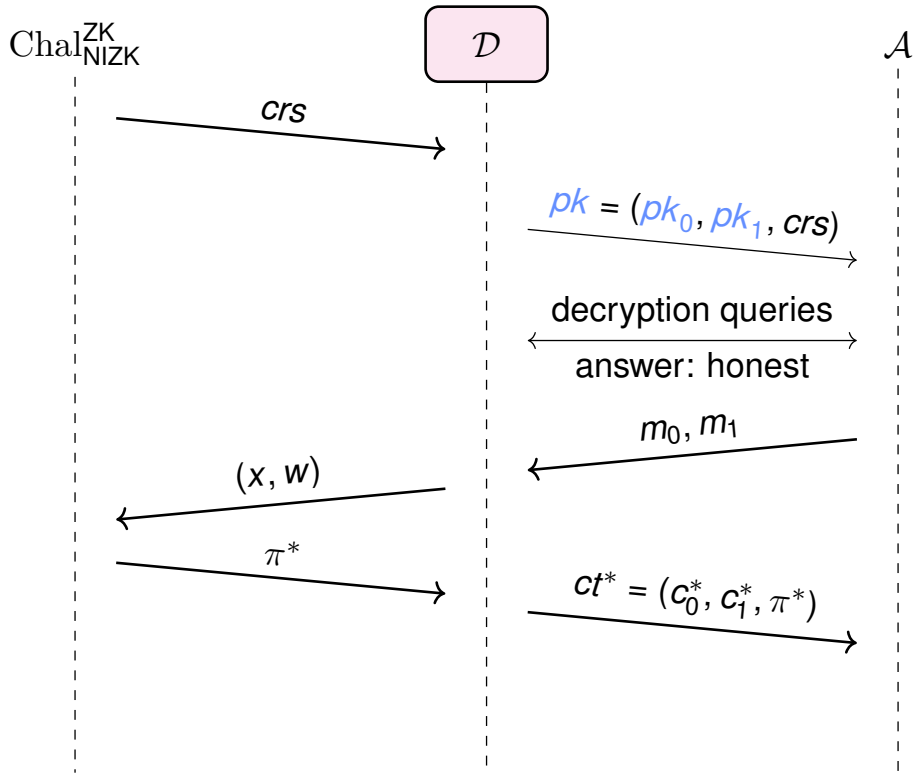
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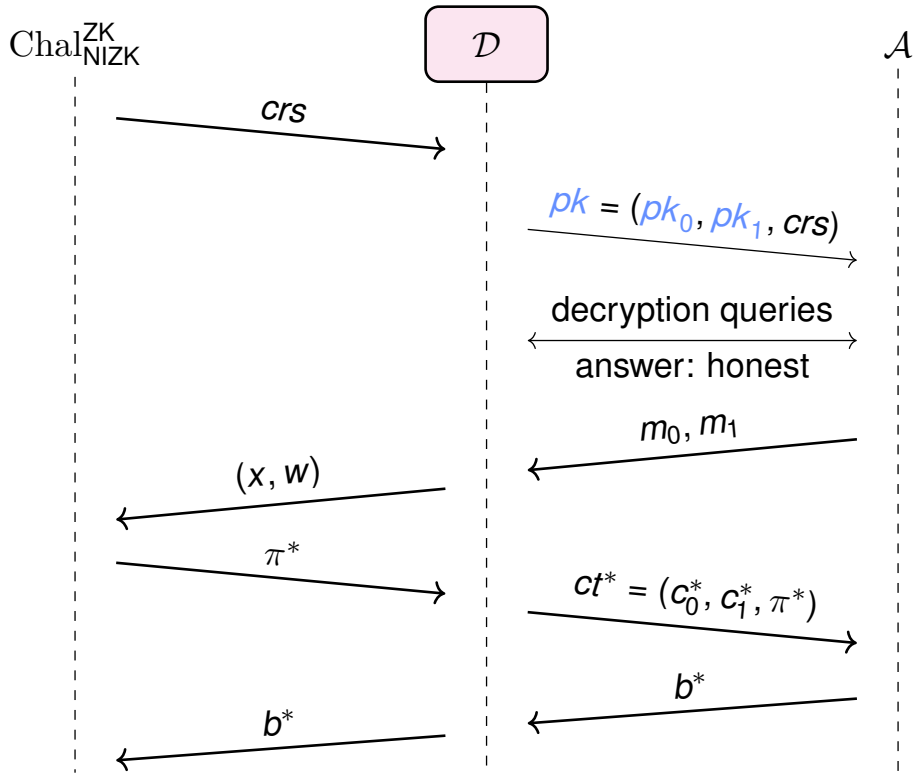


Hop $G_0 \to G_1$: the picture



$x = (pk_0, pk_1, c_0^*, c_1^*)$, $c_i^* = \text{Enc}(pk_i, m_0; r_i)$, $w = (m_0, r_0, r_1)$ — a *true* statement.

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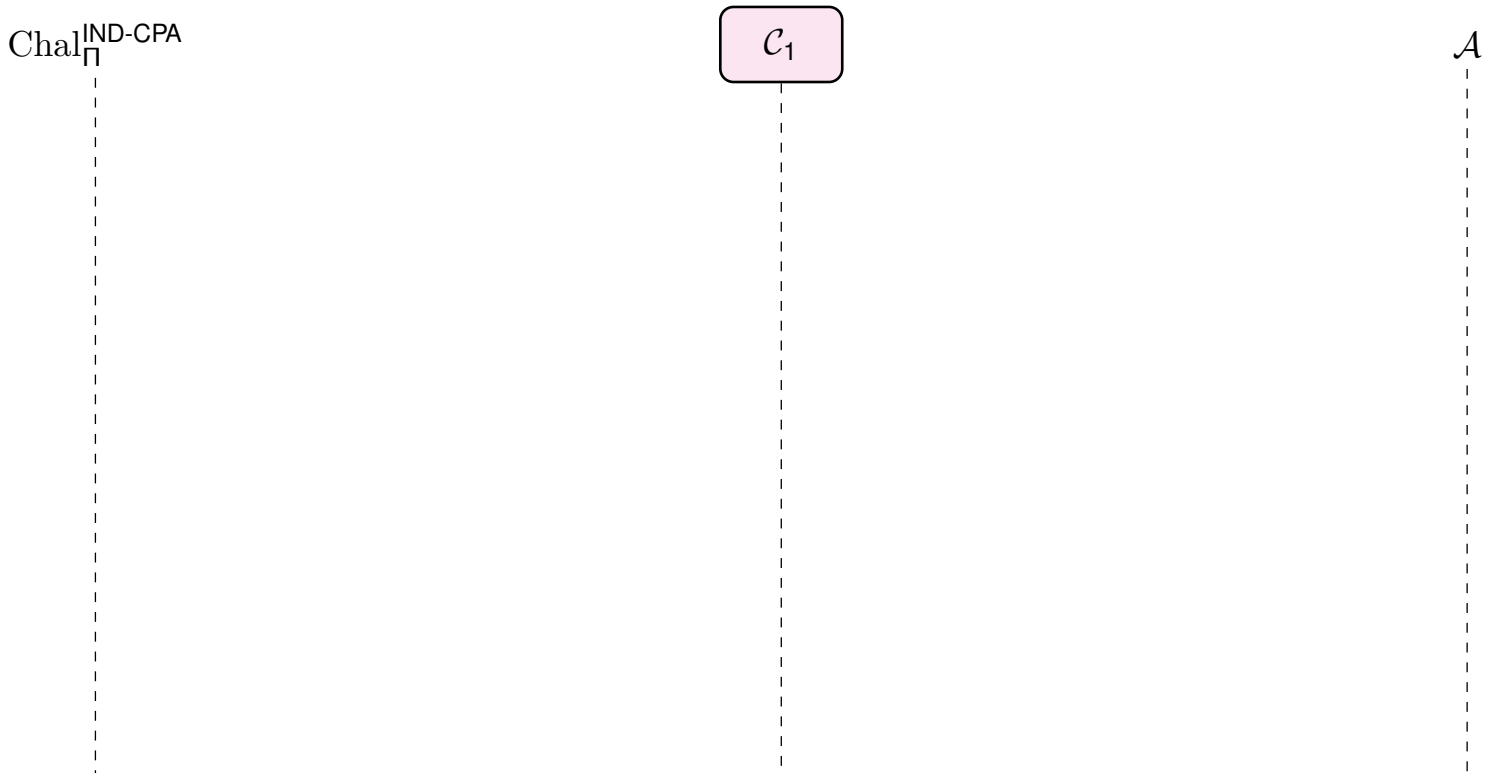
$\Pr[\text{ExpIdeal}_{\mathcal{D}} = 1] = \Pr[G_1 = 1$. Hence $\epsilon_1 = \text{Adv}^{\text{ZK}}_{\text{NIZK},\mathcal{D}}$.

Proof of Theorem 3.1: where are we?

game	c_0^* encrypts	c_1^* encrypts	*crs*, π^*	$\mathcal{O}_{\mathrm{Dec}}$ key	$\mathsf{G}_{i-1} \to \mathsf{G}_i$ via
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Only change: c_1^* encrypts m_1 instead of m_0. Reduction to IND-CPA.



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$\mathcal{C}_1$ against IND-CPA of Π

- Receive pk_1; $(pk_0, sk_0) \leftarrow \mathrm{Gen}(1^\lambda)$, $(crs, td) \leftarrow \mathcal{S}_{\mathrm{crs}}(1^\lambda)$;
- Run $\mathcal{A}(pk_0, pk_1, crs)$, answer $\mathcal{O}_{\mathrm{Dec}}$ using sk_0
- On (m_0, m_1): forward to $\mathrm{Chal}_\Pi^{\text{IND-CPA}}$, receive $c_1^* = \mathrm{Enc}(pk_1, m_b)$;
- Set $c_0^* \leftarrow \mathrm{Enc}(pk_0, m_0)$, $\pi^* \leftarrow \mathcal{S}_{\mathrm{prf}}(crs, td, (pk_0, pk_1, c_0^*, c_1^*))$
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- $b = 0$: $\mathcal{A}$ sees G_1 so $\Pr[G_1 = 1] = \Pr[\mathrm{Game}_{\Pi,\mathcal{C}_1}^{\text{IND-CPA}}(b = 0) = 1]$
- $b = 1$: $\mathcal{A}$ sees G_2 so $\Pr[G_2 = 1] = \Pr[\mathrm{Game}_{\Pi,\mathcal{C}_1}^{\text{IND-CPA}}(b = 1) = 1]$
- Hence: $\epsilon_2 = \mathrm{Adv}_{\Pi,\mathcal{C}_1}^{\text{IND-CPA}}(\lambda)$

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  - ...then $\mathcal{C}_1$ breaks IND-CPA

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G_5	m_1	m_1	simulated	sk_0	soundness
G_6	m_1	m_1	real	sk_0	zero-knowledge

Hop $G_2 \to G_3$: switching the decryption key

Only change: $\mathcal{O}_{\mathrm{Dec}}$ answers $\mathrm{Dec}(sk_1, c_1)$ instead of $\mathrm{Dec}(sk_0, c_0)$ (still after checking π).

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- BAD happens $\Rightarrow \mathcal{A}$ faked a proof (by correctness of Π)
- Intuitively, Bad violates soundness. *BUT we need to be careful!*

Hop $G_2 \rightarrow G_3$: switching the decryption key

- Goal: show that $\epsilon_3 \leq \Pr[\text{Bad in } G_2]$
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- Recall that in G_2:
  - *crs* is *simulated*
- Recall that in the soundness game
  - *crs* is *honestly generated*!
- We can't reduce directly!

Lemma 4.1: Consistency under a simulated *crs*

There are PPT $\mathcal{B}_1, \mathcal{D}_2$ such that

$$\Pr[\text{Bad}] \leq \mathsf{Adv}^{\text{sound}}_{\text{NIZK},\mathcal{B}_1}(\lambda) + \mathsf{Adv}^{\text{ZK}}_{\text{NIZK},\mathcal{D}_2}(\lambda).$$

Proof of Lemma 4.1

Proof overview:

Step 1: Honest *crs*

- Define a new game G_2'
- As G_2 with an honest *crs* $\leftarrow \mathrm{Setup}(1^\lambda)$
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Step 2: Simulated vs. honest *crs*

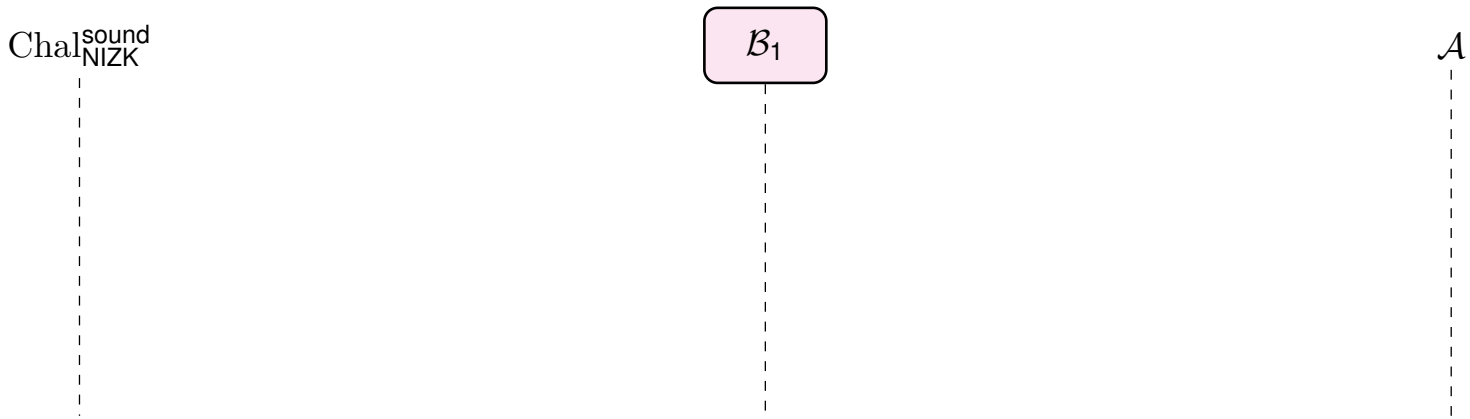
- "Bad" in G_2 and G_2' almost equally likely
- Formally

  $\left|\Pr[\text{Bad in } \mathsf{G}_2] - \Pr[\text{Bad in } \mathsf{G}_2']\right| = \mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NIZK},\mathcal{D}_2}(\lambda)$

Conclusion: Triangular inequality

Proof of Lemma 4.1 : honest *crs* (handwritten)

Let G_2' be G_2 with an honest $crs \leftarrow \mathrm{Setup}(1^\lambda)$.



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Let G_2' be G_2 with an honest $crs \leftarrow \mathrm{Setup}(1^\lambda)$. (Only the query phase matters for Bad.)

$\mathcal{B}_1$ against soundness

- receive *crs*; generate $(pk_0, sk_0), (pk_1, sk_1)$; run $\mathcal{A}$'s query phase, answering $\mathcal{O}_{\mathrm{Dec}}$ with sk_0
- if some query (c_0, c_1, π) has $\mathrm{Vfy} = 1$ and $\mathrm{Dec}(sk_0, c_0) \neq \mathrm{Dec}(sk_1, c_1)$: output $x := (pk_0, pk_1, c_0, c_1)$ and π

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- $\mathcal{B}_1$ simulates G_2' perfectly (it knows both keys).
- $\mathrm{Dec}(sk_0, c_0) \neq \mathrm{Dec}(sk_1, c_1) \Rightarrow x \notin \mathcal{L}_{\mathrm{NY}}$ (if $x \in \mathcal{L}_{\mathrm{NY}}$, correctness gives the same m on both sides).

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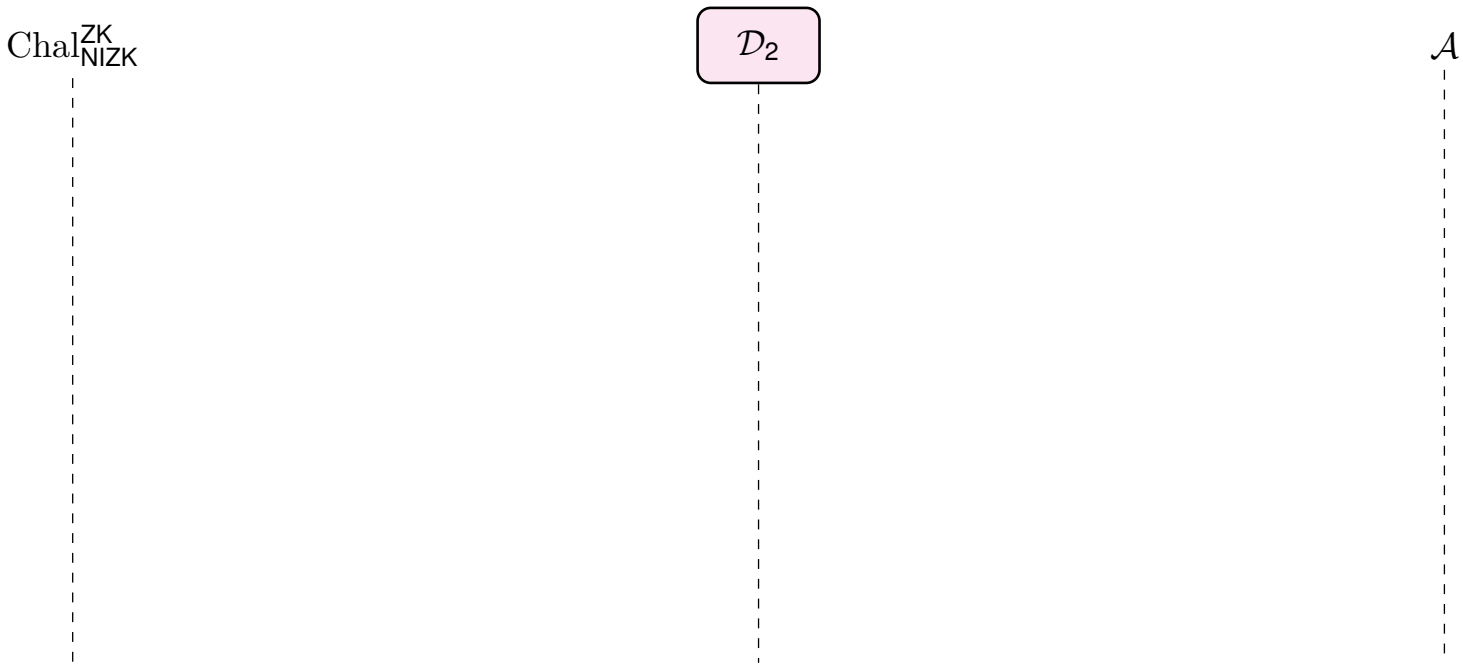
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$$\Pr[\text{Bad in } \mathsf{G}_2'] \leq \mathrm{Adv}^{\mathrm{sound}}_{\mathrm{NIZK},\mathcal{B}_1}(\lambda).$$

Proof of Lemma 4.1 : simulated vs. honest *crs*

Bad equally likely in G_2' and G_2 - reduction to ZK.



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$\mathcal{D}_2 = (\mathcal{D}_2^{(0)}, \mathcal{D}_2^{(1)})$ `against ZK`

$\mathcal{D}_2^{(0)}(crs)$: generate both key pairs; run $\mathcal{A}$'s query phase, answering $\mathcal{O}_{\mathrm{Dec}}$ with sk_0; set $\beta \coloneqq 1$ iff Bad occurred; output some true $(x, w) \in \mathcal{R}_{\mathrm{NY}}$ and $st \coloneqq \beta$

$\mathcal{D}_2^{(1)}(st, \pi)$: ignore π, output β

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- $\mathrm{ExpReal}^{\mathsf{ZK}}$: $\mathcal{A}$ runs in G_2'. $\mathrm{ExpIdeal}^{\mathsf{ZK}}$: $\mathcal{A}$ runs in G_2.
- Hence $|\Pr[\text{Bad in } \mathsf{G}_2] - \Pr[\text{Bad in } \mathsf{G}_2']| = \mathsf{Adv}^{\mathsf{ZK}}_{\mathsf{NIZK},\mathcal{D}_2}(\lambda)$; with Step 1 the Lemma 4.1 follows. □

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$\mathcal{D}_2$ never has to give $\mathcal{A}$ a proof: all queries come *before* ct^*.

Proof of Theorem 3.1: where are we?

game	c_0^* encrypts	c_1^* encrypts	*crs*, π^*	$\mathcal{O}_{\mathrm{Dec}}$ key	$\mathsf{G}_{i-1} \to \mathsf{G}_i$ via
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Hops $G_3 \to G_4 \to G_5 \to G_6$: by symmetry

- $G_3 \to G_4$: c_0^* encrypts m_1 instead of m_0.
  - Reduction $\mathcal{C}_0$ to IND-CPA of pk_0;
  - $\mathcal{C}_0$ generates (pk_1, sk_1) itself and answers $\mathcal{O}_{\mathrm{Dec}}$ with sk_1 *this is why we switched keys*.
  - $\epsilon_4 = \mathrm{Adv}^{\text{IND-CPA}}_{\Pi,\mathcal{C}_0}$.

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- $G_4 \to G_5$: switch $\mathcal{O}_{\mathrm{Dec}}$ back to sk_0.
  - Same Bad event, same Lemma 4.1
  - $\epsilon_5 \leq \mathrm{Adv}^{\text{sound}}_{\text{NIZK},\mathcal{B}_2} + \mathrm{Adv}^{\text{ZK}}_{\text{NIZK},\mathcal{D}_3}$.

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G_6 is exactly the IND-CCA1 experiment with $b = 1$.

Proof of Theorem 3.1: summing up

$$\mathsf{Adv}^{\text{IND-CCA1}}_{\Pi_{\mathrm{NY}},\mathcal{A}}(\lambda) \;\leq\; \sum_{i=1}^{6} \epsilon_i \;\leq\; 2\,\mathsf{Adv}^{\text{IND-CPA}}_{\Pi,\cdot} + 2\,\mathsf{Adv}^{\text{sound}}_{\text{NIZK},\cdot} + 4\,\mathsf{Adv}^{\text{ZK}}_{\text{NIZK},\cdot}$$

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In every game, whoever answers $\mathcal{O}_{\mathrm{Dec}}$ *knows* the key it uses; soundness guarantees that *both* keys give the same answer.

Outline

Recap: Lecture 3

Naor–Yung: IND-CCA1 security

Naor–Yung: IND-CCA2 security

Where exactly did we use IND-CCA1?

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Hope G_2 to G_3, only in Lemma 4.1

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NO! Soundness + ZK allow *malleable proofs!*

Malleable proofs give a real attack

Simulation soundness

Definition 4.1: (One-time) simulation soundness

A ZK proof system with simulator $\mathcal{S} = (\mathcal{S}_{\mathrm{crs}}, \mathcal{S}_{\mathrm{prf}})$ is *simulation sound* if for all PPT $\mathcal{A} = (\mathcal{A}_0, \mathcal{A}_1)$ the following is negligible:

$$\mathsf{Adv}^{\text{sim-sound}}_{\text{NIZK},\mathcal{A}}(\lambda) \coloneqq \Pr\left[\begin{array}{l|l} \mathrm{Vfy}(crs, x, \pi) = 1 & (crs, td) \leftarrow \mathcal{S}_{\mathrm{crs}}(1^\lambda) \\ \quad \wedge\ x \notin \mathcal{L}_{\mathcal{R}} & (x^*, st) \leftarrow \mathcal{A}_0(crs) \\ \wedge\ (x, \pi) \neq (x^*, \pi^*) & \pi^* \leftarrow \mathcal{S}_{\mathrm{prf}}(crs, td, x^*) \\ & (x, \pi) \leftarrow \mathcal{A}_1(st, \pi^*) \end{array}\right]$$

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$\mathcal{A}$ gets to see *one* simulated proof, of a statement x^* of its choice — x^* may be **false**.

Simulation soundness: remarks

- After seeing π^*, $\mathcal{A}$ must still not produce a *new* valid proof of a false statement; $(x, \pi) \neq (x^*, \pi^*)$: copying is the only thing allowed.

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- After seeing π^*, $\mathcal{A}$ must still not produce a *new* valid proof of a false statement; $(x, \pi) \neq (x^*, \pi^*)$: copying is the only thing allowed.
- Implies soundness under a simulated *crs* — which we previously argued by hand.

Simulation soundness: remarks

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- *One-time* (one simulated proof) is what Naor–Yung needs: one challenge ciphertext.
- Re-randomisable proofs are **not** simulation sound: pick a false x^*, maul π^*.
- *Existence:* any NIZK for NP $\rightarrow$ one-time simulation-sound NIZK for NP [Sahai99, Lin03].

  $\Rightarrow$ no new assumption. Efficient constructions: e.g. [Gro06] from pairings.

Naor-Yung: IND-CCA2 security

Construction: *unchanged*. Π_{NY} as before, instantiated with a simulation-sound NIZK.

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History.

- [NY90] proved IND-CCA1 with a plain NIZK.
- [DDN91] achieved IND-CCA2 with a much more involved construction.
- [Sahai99] observed that the Naor–Yung scheme itself is IND-CCA2 once the NIZK is simulation sound.

Proof of Theorem 4.2: the games

game	c_0^* encrypts	c_1^* encrypts	*crs*, π^*	$\mathcal{O}_{\mathrm{Dec}}$ key	$\mathsf{G}_{i-1} \to \mathsf{G}_i$ via
G_0	m_0	m_0	real	sk_0	—
G_1	m_0	m_0	simulated	sk_0	zero-knowledge
G_2	m_0	m_1	simulated	sk_0	IND-CPA of pk_1
G_3	m_0	m_1	simulated	sk_1	**sim. soundness**
G_4	m_1	m_1	simulated	sk_1	IND-CPA of pk_0
G_5	m_1	m_1	simulated	sk_0	**sim. soundness**
G_6	m_1	m_1	real	sk_0	zero-knowledge

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Unchanged hops. ZK: $\mathcal{D}$ knows both keys, answers all queries. IND-CPA: $\mathcal{B}$ knows the key $\mathcal{O}_{\mathrm{Dec}}$ uses in both games. ✓

Hop $G_2 \to G_3$: simulation soundness

Define Bad event in G_2 (same as before):

A query $(c_0, c_1, \pi) \neq ct^*$ with $\mathrm{Vfy} = 1$ and
$\mathrm{Dec}(sk_0, c_0) \neq \mathrm{Dec}(sk_1, c_1)$

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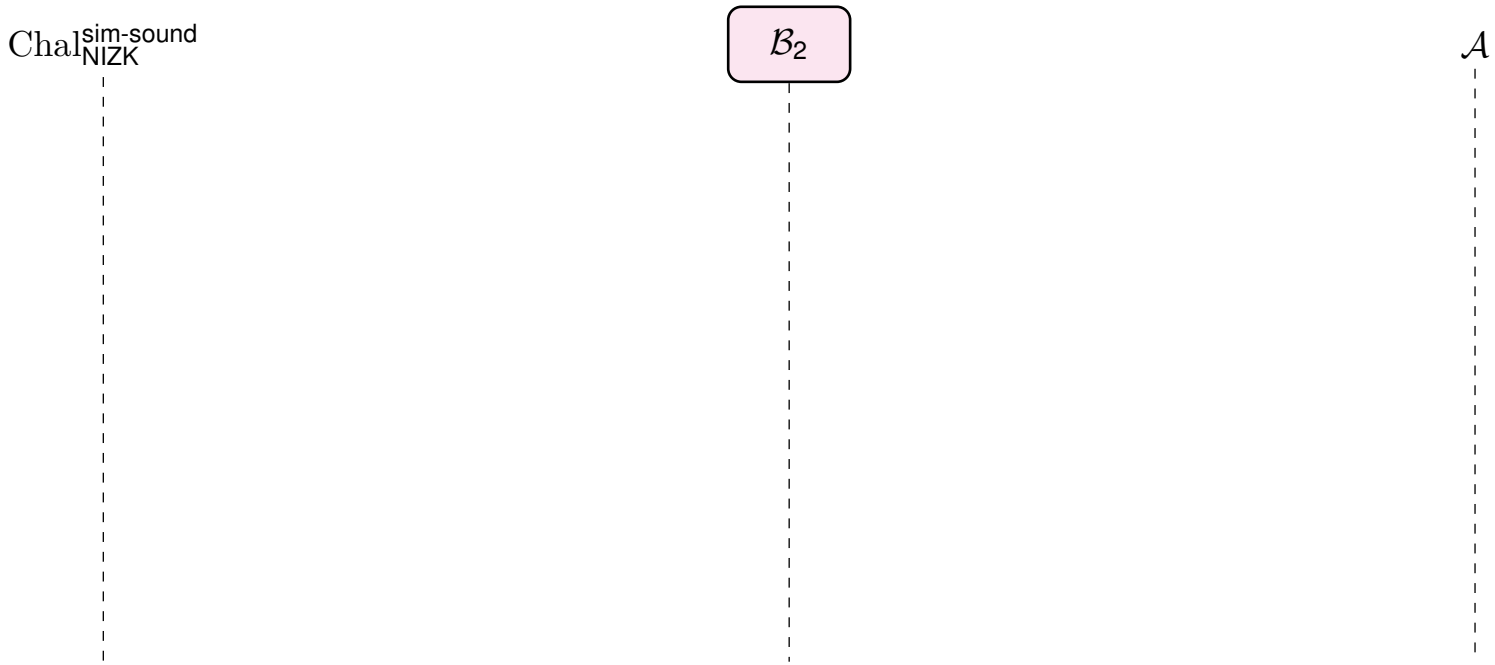
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To prove:

$$\Pr[\text{Bad in } G_2] \leq \mathrm{Adv}^{\text{sim-sound}}_{\text{NIZK},\mathcal{B}_2}$$

Hop $G_2 \to G_3$: simulation soundness (handwritten)



Hop $G_2 \to G_3$: simulation soundness

$\mathcal{B}_2 = (\mathcal{B}_{2,0}, \mathcal{B}_{2,1})$ against simulation soundness

$\mathcal{B}_{2,0}(crs)$:

- $(pk_0, sk_0), (pk_1, sk_1) \leftarrow \mathrm{Gen}(1^\lambda)$; run $\mathcal{A}(pk_0, pk_1, crs)$
- answer $\mathcal{O}_{\mathrm{Dec}}$ with sk_0, watch for Bad
- on (m_0, m_1):
  - $c_0^* \leftarrow \mathrm{Enc}(pk_0, m_0)$, $c_1^* \leftarrow \mathrm{Enc}(pk_1, m_1)$;
  - output $x^* := (pk_0, pk_1, c_0^*, c_1^*)$

$\mathcal{B}_{2,1}(st, \pi^*)$:

- give $ct^* = (c_0^*, c_1^*, \pi^*)$ to $\mathcal{A}$;
- keep answering $\mathcal{O}_{\mathrm{Dec}}$ with sk_0, watch for Bad
- on Bad at query (c_0, c_1, π): output $x := (pk_0, pk_1, c_0, c_1)$ and π

Hop $G_2 \to G_3$: simulation soundness (cont.)

- $\mathcal{B}_2$ simulates G_2 perfectly: *crs* and π^* are simulated there too.
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$\epsilon_3 \leq \Pr[\text{Bad in } G_2] \leq \mathrm{Adv}^{\text{sim-sound}}_{\text{NIZK},\mathcal{B}_2}(\lambda)$. Same for ϵ_5.

Proof of Theorem 4.2: summing up

$$\begin{aligned}\mathsf{Adv}^{\text{IND-CCA2}}_{\Pi_{\mathrm{NY}},\mathcal{A}}(\lambda) &\leq 2 \cdot \mathsf{Adv}^{\text{IND-CPA}}_{\Pi,\cdot} + 2 \cdot \mathsf{Adv}^{\text{sim-sound}}_{\text{NIZK},\cdot} + 2 \cdot \mathsf{Adv}^{\text{ZK}}_{\text{NIZK},\cdot} \\ &\leq \mathsf{negl}(\lambda). \qquad \square\end{aligned}$$

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Same scheme, same proof skeleton. Upgrading soundness $\rightarrow$ simulation soundness is all it takes to go from IND-CCA1 to IND-CCA2.

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Next: Cramer–Shoup — same “publicly checkable consistency” idea, NIZK replaced by an efficient *designated-verifier* proof.

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